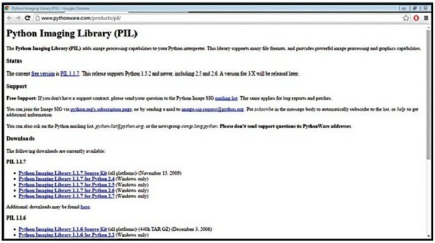

Fig. 3: Downloading Python Imaging Library for Python 2.7

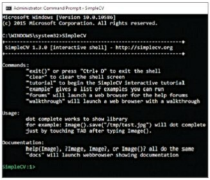

Fig. 4: SimpleCV command window

Fig. 5: USB camera detected

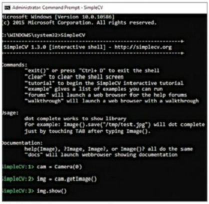

Fig. 6: Camera commands

pleCV and all the dependencies one by one

3. Download Python Imaging Library (PIL) for Python 2.7 from http://www.pythonware.com/products/pil/ as shown in Fig. 3 and install it

Test SimpleCV. Run command prompt (as administrator), type in SimpleCV and press Enter. SimpleCV shell should appear now for accepting commands as shown in Fig. 4.

As the recent version of SimpleCV doesn't work well with the latest version of ipython, you need to uninstall the latest version of ipython and install an older one using below commands under C:\Windows\Systems32:

```
> pip uninstall ipython
> pip install ipython==4.2.0
```

(Before you run these commands in command prompt, you need 'pip' already installed. You can install it by using > python -m pip install -U pip command.)

Setup for USB camera

You need to complete the setup before proceeding further. The setup is simple, requiring just a USB camera attached to a laptop or PC. This project used a regular USB webcam, which is easily available in the market. Plug in the camera and check whether it is detected in Device Manager (Fig. 5). The camera should be mounted such that it does not move and captures the whole tablet pack underneath. No additional lighting is required. The camera effectively captures images in low lighting as well.

Test camera. Run the command prompt as an administrator, type SimpleCV and press 'Enter.'

Once you get SimpleCV shell, type the following commands to test the camera (Fig. 6):

```
>cam = Camera(0)
>img = cam.getImage()
>img.show()
```

If you get images from the USB camera, your camera has been detected as Camera(0) by the operating system. Otherwise, change Camera(0) to Camera(1) in above commands and try again. The reference camera is used in the software.

Python IDE

Python IDE, which comes with Python installation, is used to write, test and debug Python programs. When Python is run, it opens up a Python Shell window as shown in Fig. 7.

You can run it directly from the installed applications by clicking its logo. Or, write IDLE in the search tab on the Windows start page to get the option to be clicked. Python is an interpreted language, so you can immediately start writing the commands and press 'Enter.' The commands get executed when you press 'Enter.' Test by typing 2+2 and then press 'Enter.' You should get four as the answer in the next line.

Create new program

Run Python IDLE→File→New File

Now you can start writing the program in the IDLE. The complete program is provided in Quality